Bilingual Text Summarizer

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What is Text Summarization?

Text summarization condenses large documents into shorter versions while retaining key information.

Helps users save **time and effort** in reading long texts.

Two main approaches:

Extractive Summarization

Selects important sentences/phrases from the original text.

Abstractive Summarization

Generates new sentences conveying the same meaning (more human-like).

AI models like **BART-Large** enable **better understanding and rephrasing**.

Applications: News, research papers, legal documents, educational content, and content simplification.

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Problem Statement

Time-Consuming

Reading long documents is **time-consuming**.

Inefficient Process

Manual summarization is **inefficient and inconsistent**.

Language Limitations

Existing summarizers often support **only a single language**, limiting accessibility.

The Solution Needed

Need for a **bilingual AI-based summarizer** that is flexible and user-friendly.

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Project Objectives

01

Build a BART-Large based AI summarizer for bilingual support.

02

Generate human-like, coherent summaries.

03

Provide multiple summarization modes for different user needs.

04

Allow structured summaries with user-defined paragraphs and sentences.

05

Enable translation between English and Hindi for input and output.

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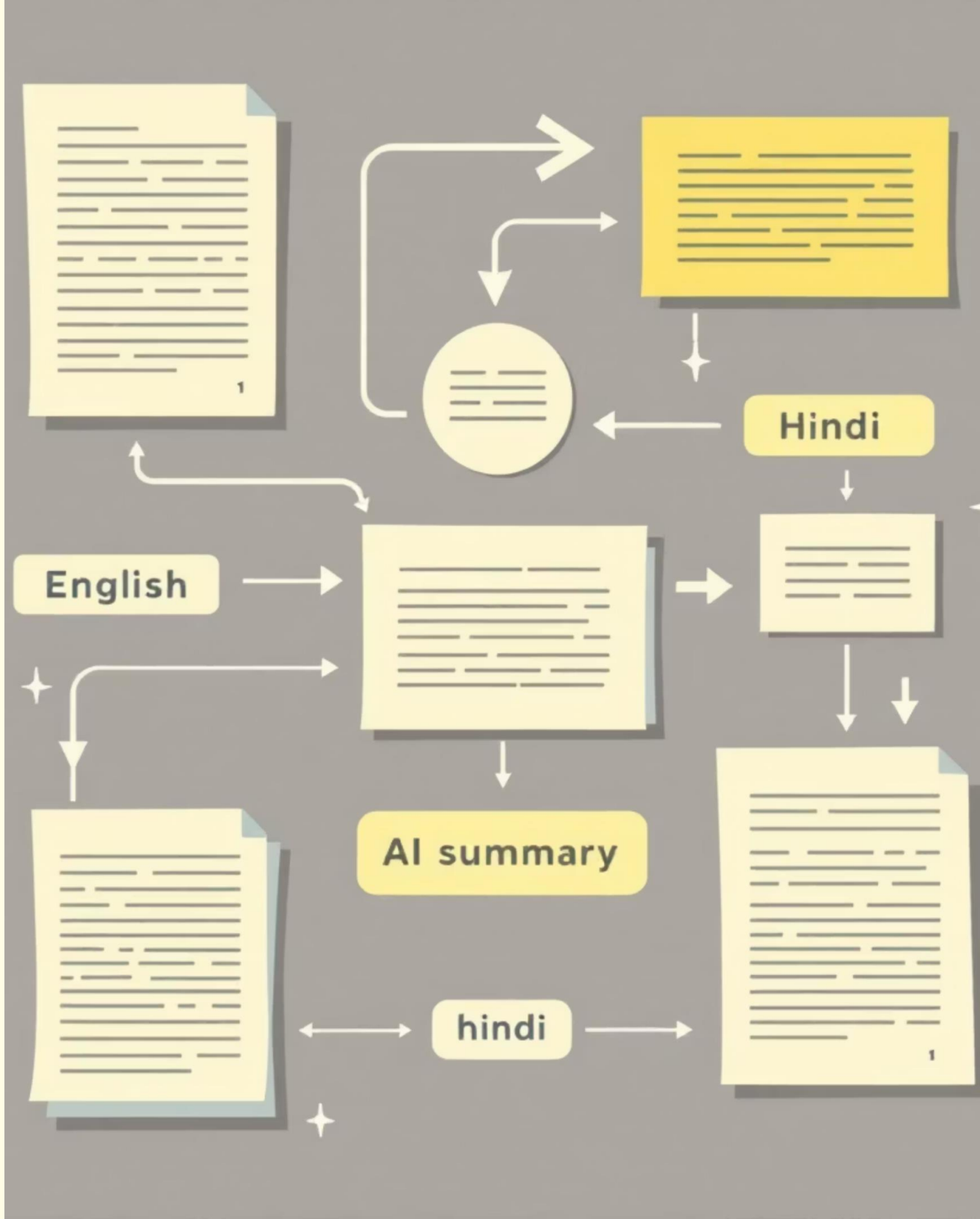
Bilingual Summarization Summarization

- Supports **English and Hindi** input and output.
- Uses **Google Translate API** for translation.
- Input options: English, Hindi, or Auto-detect language.

- Output options: English or Hindi summary.
- Ensures **meaning preservation across languages**.

- Useful for research, education, journalism, and professional documentation.



System Architecture

Flow:

Input

User enters text manually or loads from file.

Language Detection

Input language is detected or selected by user.

Translation

If Hindi, text is translated to English for summarization.

Mode Selection

User selects **summarization mode** (single, paragraph-wise, long, quick, detailed, structured).

AI Processing

BART-Large model generates the summary.

Output Translation

Summary is translated to output language if required.

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Technologies Used

Python

Main programming language.

PyTorch

Deep learning framework used to load and run the BART-Large model. It manages tensors, model weights, and CPU/GPU processing efficiently.

Transformers Library

Provides the BART-Large (facebook/bart-large-cnn) model and tokenizer. It handles text-to-token conversion and generates high-quality abstractive summaries.

NLTK

Used for splitting text into sentences, cleaning text, and creating chunks for long document summarization.

Google Translate API

Adds bilingual capability by translating text between English and Hindi, making the system usable for more users.

TQDM & Logging

TQDM and Logging help improve the program by showing progress during long tasks and recording important events or errors for easy debugging.

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Key Features

1

Abstractive Summaries

Generates **abstractive summaries** (human-like and coherent).

2

Document Flexibility

Handles **short and long documents** efficiently.

3

Multiple Modes

Offers **multiple summarization modes**: single, paragraph-wise, long, quick, detailed, structured.

4

Custom Structure

Custom structured summaries: user can set number of paragraphs and sentences per paragraph.

5

Bilingual Support

Bilingual support: English & Hindi input/output.

6

Statistics

Provides **summary statistics** for evaluation.

7

User-Friendly

User-friendly **interactive interface**.

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User Options

Input Options

- **Input:** Manual text entry or file upload.
- **Languages:** English, Hindi, or auto-detect input; English or Hindi output.
- **Modes:** Single, Paragraph-wise, Long, Quick, Detailed, Structured.
- **Structured mode:** User defines number of paragraphs and sentences per paragraph.

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Output

```
SUMMARY (FULL)
----------------------------------------------------------------------
The definition of terrorism is complex and nuanced, with no universality. It has been defined differently by some differ
ent groups of reporters, including military intelligence. The broadest term generally defines intentional violence commi
tted against non-combatants during the course of war or peacetime. Modern terrorists aim to instill fear among their tar
get's citizens so that they will make policy changes or seek greater security.
----------------------------------------------------------------------
Time taken: 18.81 seconds

STATISTICS (approximate)
----------------------------------------------------------------------
Original: 175 words, 7 sentences
Summary:  70 words, 4 sentences
Reduction: 60.0%
----------------------------------------------------------------------

FULL ORIGINAL TEXT
----------------------------------------------------------------------
Terrorism, in its broadest sense, is the use of violence against non-combatants to achieve political or ideological aims
.[1] The term is used in this regard primarily to refer to intentional violence during peacetime or in the context of wa
r against non-combatants.[2] There are various different definitions of terrorism, with no universal agreement about it.
[3][4][5] Different definitions of terrorism emphasize its randomness, its aim to instill fear, and its broader impact b
eyond its immediate victims.[1]

Modern terrorism, evolving from earlier iterations, employs various tactics to pursue political goals, often leveraging
fear as a strategic tool to influence decision makers. By targeting densely populated public areas such as transportatio
n hubs, airports, shopping centers, tourist attractions, and nightlife venues, terrorists aim to instill widespread inse
curity, prompting policy changes through psychological manipulation and undermining confidence in security measures.[6]
----------------------------------------------------------------------

Summarize another? (y/n):
```

Conclusion & Future Scope

Conclusion:

- Developed a **bilingual text summarizer** using **BART-Large**.
- Supports **flexible summarization modes** and structured output.
- Provides **English and Hindi summaries** while maintaining readability.

Future Scope:

- Add support for **more languages** (Spanish, French, etc.).
- Integrate with **web/mobile applications** for real-time use.
- Improve **translation accuracy** and AI understanding of context.
- Enable **voice input/output** for accessibility.
- Explore **real-time summarization** for live news or lectures.

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